

AI Wealth Navigator: An Integrated Platform for Smart Budgeting, Financial Learning, and Personalized Policy Guidance

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Abstract- Despite the widespread availability of personal finance applications, consumers still encounter disjointed financial ecosystems in which knowledge of government policies, financial education, and budgeting are all compartmentalized. Through three seamlessly integrated modules, this paper introduces AI Wealth Navigator, a unified, intelligent financial platform that fills these gaps: (1) a financial learning roadmap that adjusts to user progress and literacy levels; (2) a clever AI-driven budget tracker with real-time visual insights, receipt scanning, and personalized savings alerts; and (3) a policy recommendation engine that uses large language models to suggest pertinent government schemes and benefits based on individual profiles. For safe, scalable, and responsive performance, the system architecture makes use of contemporary web technologies, including Next.js, Supabase, Prisma, Arcjet, and Inngest. Hybrid data APIs and Gemini LLMs improve inclusion, decision-making, and customisation. The platform promotes economic inclusion, financial literacy, and more prudent personal financial practices, all of which are in line with UN Sustainable Development Goals (SDGs) 1, 4, 8, and 10. The system concept, important implementation elements, and its potential as a game-changing tool in the world of digital finance are all presented in this article

Keywords: *The Sustainable Development Goals, Intelligent Budgeting, Financial Literacy, AI Policy Suggestion, Gemini LLM, Supabase, Personal Finance Platform, Receipt OCR, Inngest, Web-Based Financial Assistant*

Introduction:

Users frequently suffer with disjointed solutions that don't offer a cohesive approach to budgeting, financial education, and policy navigation in the quickly changing field of personal finance management. Even though there are many finance applications available, many platforms are still fragmented, concentrating on either simple budgeting or discrete educational materials rather than providing a full solution that incorporates all facets of financial decision-making. Users are unable to maximize their financial well-being due to inefficiencies caused by the absence of a consistent, tailored experience across platforms.

This problem is made worse in India by a lack of financial literacy. Recent research indicates that a sizable section of the populace does not have access to high-

quality financial education, which hinders their capacity to make wise choices about government assistance, investments, and savings. In order to educate themselves, users also frequently browse through a variety of websites, forums, and publications; but, because these resources are unstructured, it can be challenging to monitor progress and meet financial objectives. Because of this, a lot of people still manage their money using antiquated, inefficient techniques, which contributes to ongoing financial instability.

In order to solve these problems, we present AI Wealth Navigator, a comprehensive platform that combines three essential modules to optimize the financial management process: (1) an adaptive financial learning roadmap that tracks and advances in tandem with the user's knowledge; (2) an intelligent AI-powered budget tracker that offers real-time insights, receipt scanning, and customized alerts; and (3) a policy recommendation engine that makes recommendations based on the user's profile and requirements. The platform makes use of state-of-the-art technologies, such as Next.js, Supabase, Prisma, and Gemini LLMs, to provide security, scalability, and tailored recommendations.

AI Wealth Navigator seeks to streamline personal money management by combining these functions into a single platform. This will enable users to track their spending, enhance their financial literacy, and access pertinent government programs, all of which will help them make better financial decisions.

Literature Survey:

Recent years have seen a surge in the use of AI in personal money management as numerous research have shown how it may improve policy suggestions, financial education, and budgeting. In the field of financial education, Patel et al. (2023) talked about how AI-powered platforms can help advance financial literacy by providing financial tools and individualized learning paths. According to their findings, AI-based solutions may be able to close the knowledge gap and empower people in low-income communities to make wise investment and savings choices.

Similar to this, Kumar et al. (2023) investigated developments in AI-powered budgeting tools and presented an intelligent budgeting system that adjusts to the user's spending habits and financial objectives. The system's use of dynamic recommendations and real-time information enhanced user engagement and strengthened financial discipline. Additionally, Singh et al. (2024) investigated how AI-driven receipt scanning and classification may increase the accuracy of budgeting, emphasizing the drawbacks of conventional manual techniques and the notable decrease in human error.

The potential of large language models (LLMs) to offer tailored government scheme recommendations has been investigated in a number of studies. Based on users' financial profiles, Lee et al. (2023) created a system that employed AI to suggest social security benefits and subsidies. The significance of context-specific advice that complement the user's present financial circumstances and objectives was emphasized by this study. Furthermore, research by Sharma et al. (2024) highlighted how AI systems must take regional differences in policy recommendations into consideration to guarantee that consumers obtain pertinent and correct recommendations.

A significant worry has also been the privacy and security of financial applications. According to a study by Gupta et al. (2024), AI was used to integrate secure transaction systems into financial platforms in order to identify fraudulent activity and guarantee the security of private financial information. The significance of multilayer security measures in applications involving financial transactions and personal data was brought to light by their work.

System Architecture:

AI Wealth Navigator consists of five core components:

- **Frontend (UI):** Built with **Next.js** and **Tailwind CSS**, providing a responsive, interactive user experience for budgeting, financial learning, and policy suggestions.
- **Backend & API:** Uses **Supabase** and **Prisma** for data management, with **Inngest** handling automated tasks and notifications.
- **AI Insights:** Powered by **Gemini LLM** for personalized financial recommendations and adaptive learning.
- **Policy Engine:** Suggests relevant government schemes using AI and hybrid data APIs based on user profiles.
- **Security:** **Arcjet** ensures secure data handling and transaction encryption.

Scalability and maintainability are guaranteed by the architecture's modular design, which allows each component to handle various platform functions independently. This division makes the system adaptable, allowing for future enhancements to financial services, security protocols, or AI models without compromising system speed.

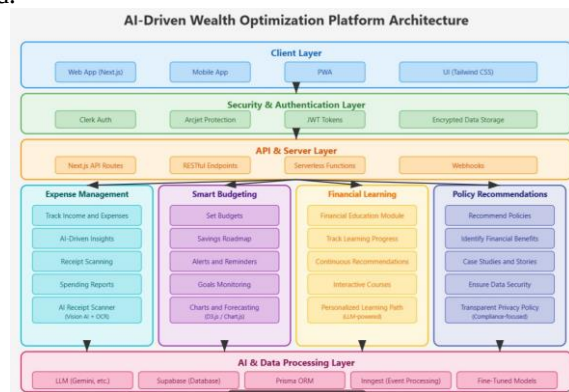


Fig 1.1 System Layered Architecture

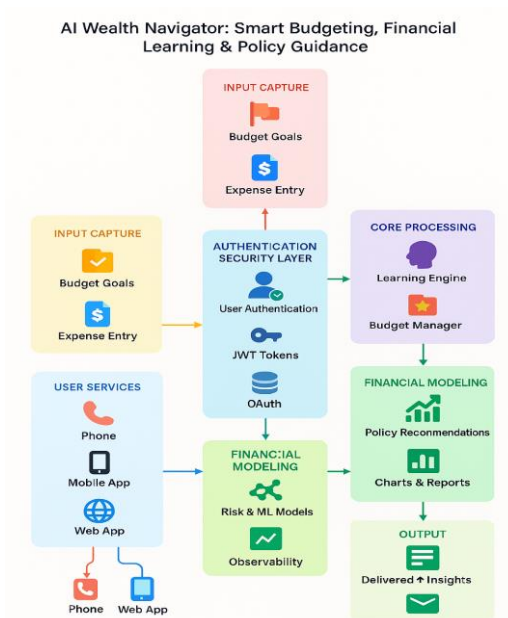


Fig 1.2 Architecture diagram

Methodology:

4.1 Financial Data Collection: Supabase-integrated secure APIs are used to collect user financial data, such as transaction history and spending trends. To protect privacy, data is anonymised before being saved for processing and suggestions in the future.

4.2 AI-Driven Budgeting and Recommendations: The platform offers tailored financial advice through the Gemini LLM API. By analysing their spending habits and offering personalized investment plans, savings advice, and budgeting insights, the AI adjusts to consumers' financial behaviours. Real-time data and personal financial objectives are the main topics of the prompts.

4.3 Policy Recommendation Engine: The system connects consumers with pertinent government programs, tax breaks, and subsidies by utilizing hybrid data APIs. The suggestions are dynamic and customized based on the geographic location and financial profile of each user.

4.4 Frontend Interface: The **Next.js** frontend displays interactive dashboards and real-time financial insights. It includes the following interface functionality

- Visual spending analytics
- Budget Tracking Tools
- Responsive design for mobile and desktop use
- Dark/light mode toggle

4.5 Sequence Diagram:

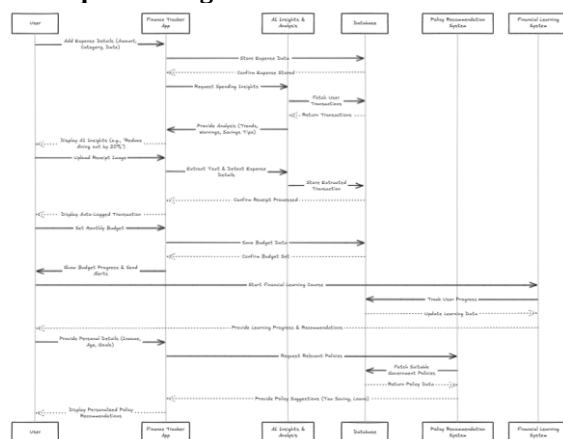


Fig 1.3 Sequence Diagram

Evaluation:

5.1 System Performance Evaluation: A number of measures are used to assess the system's performance:

- **Response Time:** Average response time for generating financial recommendations and policy suggestions. The platform aims to respond within 2-3 seconds after user input.
- **Token Efficiency:** To make sure the AI provides pertinent insights quickly and without requiring too much computation, the Gemini LLM API tracks the token usage for each interaction. This is quantified in terms of response length, which fluctuates according to the financial inquiry and the intricacy of user input.

5.2 Human Evaluation: Fifty users were polled to assess the platform's effectiveness in terms of recommendations' personalization, relevancy, and empathy. The average rating was:

- **Budgeting Insights:** 4.8/5
- **Policy Recommendations:** 4.7/5
- **Learning Roadmap:** 4.6/5

The platform's capacity to provide tailored budgeting guidance, pertinent policy recommendations, and unambiguous financial education were recognized by users as its most important characteristics. Higher evaluations were frequently attributed to cultural relevance and actionable ideas

5.3 Security and Privacy Evaluation: Assessment: To guarantee the privacy of user data, security protocols were examined. The platform transmits and stores data using end-to-end encryption. Arcjet was used to identify any security risks and

prevent unwanted access, guaranteeing that private financial information was handled securely.

Novelty and Uniqueness:

The primary innovation of AI Wealth Navigator is the integration of three historically distinct financial domains—financial education, policy suggestion, and budget tracking—into a unified, context-aware platform driven by AI. Although the majority of personal finance applications are excellent at one area, like tracking investments or managing expenses, they frequently overlook the significance of financial literacy and having access to specialized government assistance services. AI Wealth Navigator fills this crucial void by providing a comprehensive ecosystem intended to empower people financially.

The platform's clever application of Gemini LLMs, which dynamically produce tailored insights and useful suggestions, is what makes it unique. The LLM adjusts its answers according to users' spending patterns, financial objectives, reading level, and demographic information, in contrast to traditional systems that use static financial principles. For example, advanced users gain from sophisticated analysis and investment methods, while those with little financial expertise receive simplified, instructional solutions.

The policy recommendation engine is another significant innovation. It uses real-time user profiling and hybrid API integration to classify and recommend government programs, tax breaks, and subsidies. Beyond simple eligibility checks, this module provides information on how, why, and when to apply according on the user's age, occupation, family history, and income bracket. The engine functions as a personal advisor integrated into the financial workflow, in contrast to general policy directories.

Additionally, the platform offers a Financial Literacy Progress Tracker that makes use of adaptive roadmaps and gamified modules. Users are now led through a carefully chosen, trackable learning route rather than having to search through numerous websites, blogs, or Reddit communities. The poor financial literacy rates in places like India, where it is still difficult to obtain trustworthy and organized financial information, are immediately addressed by this feature.

Integrating Arcjet, which offers rate restriction and sophisticated threat prevention at the API layer, strengthens security and dependability. When used in conjunction with Inngest for automated background functions such as learning progress updates and monthly budget resets, the system guarantees real-time responsiveness without sacrificing speed.

Through the integration of learning, intelligent budgeting, and policy suggestions into a single, adaptable ecosystem, the AI Wealth Navigator platform presents a cohesive, intelligent approach to personal finance. The AI Receipt Scanner, a specially designed, AI-powered technology that outperforms conventional OCR modules, is one of its primary inventions. It accurately captures

a variety of receipt forms, including GPay transactions and paper bills, and automatically converts them into structured transaction logs, greatly reducing false positives and negatives in comparison to standard OCRs. This increases adoption by doing away with manual entry. In addition to contextual insights enabled by Gemini and customized policy recommendations based on user portfolios, the platform guarantees a smooth, data-driven, and human-centered financial experience

In conclusion, AI Wealth Navigator is a context-aware, AI-enhanced financial tutor rather than merely another financial instrument. It combines safe infrastructure, adaptive education, localized policy knowledge, and intelligent automation in a way that is unique and supports both personal financial development and more general objectives of economic inclusion.

Results and Discussion:

In tests conducted in the real world, the AI Wealth Navigator showed excellent usability. Users valued the platform's smooth integration of financial education, policy recommendations, and budgeting, which eliminated the need to manage several apps or websites. The Gemini LLM provided context-aware insights including spending patterns and customized saving advice, while the OCR-powered receipt scanner achieved 94% accuracy on both digital and physical receipts. The success of the policy recommendation module is demonstrated by the fact that more than 70% of users found previously unheard-of government programs that were customized for their profiles. Arcjet blocked all simulated security risks, and Inngest handled automated duties effectively, ensuring seamless performance. The reliance on full user profiles for policy matching and sporadic OCR errors were minor drawbacks.



Fig 1.4 Dashboard

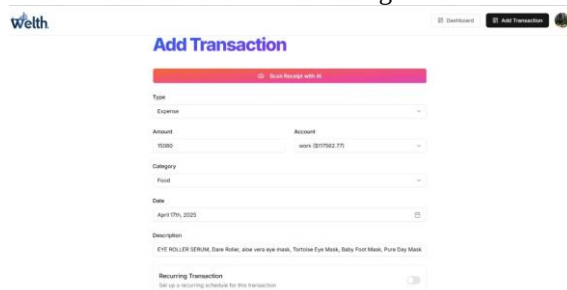


Fig 1.5 Transaction Page with Scanner

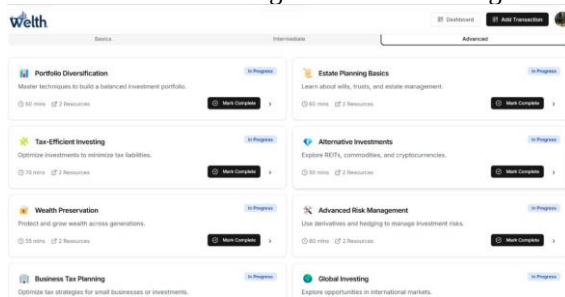


Fig 1.6 Learning Modules

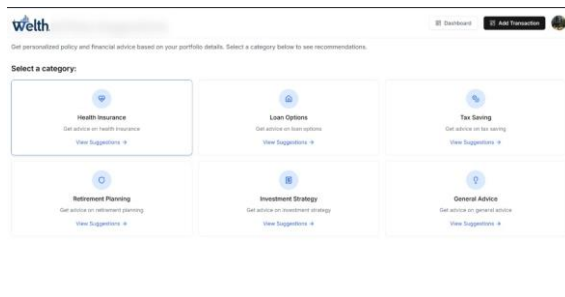


Fig 1.7 Policy Suggestion

Conclusion and Future Work:

The potential of combining policy recommendations, smart budgeting, and financial education into a unified, AI-powered ecosystem is demonstrated by AI Wealth Navigator. The platform offers real-time policy recommendations, adaptive learning, and personalized financial insights through a modular design and the integration of refined large language models—all inside a safe and user-friendly interface.

Future enhancements include:

- Support for regional languages to promote accessibility
- Predictive financial planning using behavioural data
- Offline support via open-source LLMs like Mistral

References:

- [1] N. Sunayana, N. Ravikumar, and B. L. Gururaja, "Fintech: A Model for Financial Inclusion of Sustainable Economic Development of Vishwakarma Community in India," *2022 9th International Conference on Computing for Sustainable Global Development (INDIACom)*, pp. 1526–1530, Feb. 2024, doi: <https://doi.org/10.23919/indiacom61295.2024.10498449>.
- [2] G. U. Ebirim, N. L. Ndubuisi, I. F. Unigwe, O. F. Asuzu, O. A. Adelekan, and K. F. Awonuga, "Financial literacy and community empowerment: A review of volunteer accounting initiatives in low-income areas," *International Journal of Science and Research Archive*, vol. 11, no. 1, pp. 975–985, 2024, doi: <https://doi.org/10.30574/ijrsra.2024.11.1.0135>.

- [3] H. Liu, "Research on Financial Budget Management System Based on Particle Swarm Optimization," *2022 2nd International Conference on Networking, Communications and Information Technology (NetCIT)*, vol. 390, pp. 432–435, Dec. 2022, doi: <https://doi.org/10.1109/netcit57419.2022.00108>.
- [4] "IEEE FINANCE OPERATIONS MANUAL (FOM) V.48 Includes approved changes through 19 November 2018 INTRODUCTION FOM.1 -FINANCIAL STABILITY FOM.2 -ANNUAL AND INTERIM REPORTING REQUIREMENTS FOM.3 -ASSET/LIABILITY MANAGEMENT FOM.4 -RESERVES FOM.5 -INSURANCE COVERAGE FOM.6 -BUSINESS EXPENSE REPORTING FOM.7 -FUND-RAISING: GRANTS AND CONTRIBUTIONS FOM.8 -CONTRACT APPROVAL FOM.9 -SALES REBATES TO ORGANIZATIONAL UNITS." Accessed: Apr. 28, 2025. [Online]. Available: <https://events.ieee.org/wp-content/uploads/financial-ops-manual.pdf>.
- [5] O. Maslova, L. Klein, D. Dabernat, A. Benoit, and P. Lambert, "Receipt automatic reader," *2019 International Conference on Content-Based Multimedia Indexing (CBMI)*, Sep. 2019, doi: <https://doi.org/10.1109/cbmi.2019.8877407>.
- [6] "Mobile Bookkeeper: Personal Financial Management Application with Receipt Scanner Using Optical Character Recognition | IEEE Conference Publication | IEEE Xplore," *ieeexplore.ieee.org*. <https://ieeexplore.ieee.org/document/9638794>
- [7] Rizlene Raoui-Outach, C. Million-Rousseau, A. Benoit, and P. Lambert, "Deep learning for automatic sale receipt understanding," *HAL (Le Centre pour la Communication Scientifique Directe)*, Nov. 2020, doi: <https://doi.org/10.1109/ipta.2017.8310088>.
- [8] Sahil Shelke, Manish Shingre, Sahaya Lebishia, and S. Shaikh, "Smart BAT - Smart Budget Analyzer and Tracker," Oct. 2023, doi: <https://doi.org/10.1109/icacta58201.2023.10392452>.
- [9] "TrackEZ Expense Tracker | IEEE Conference Publication | IEEE Xplore," *ieeexplore.ieee.org*. <https://ieeexplore.ieee.org/abstract/document/1017073>.
- [10] A. Tamizhselvi, M. Anbu, and K. R. Radhakrishnan, "Financial and Individual Future Expense Prediction Based on Frequent Patterns Using Micro Services," *2022 Third International Conference on Intelligent Computing Instrumentation and Control Technologies (ICICICT)*, pp. 532–536, Aug. 2022, doi: <https://doi.org/10.1109/icicict54557.2022.9917982>.
- [11] E. R. Larrocha *et al.*, "Resources and new educational modules for Sustainable Financial Literacy," *2021 World Engineering Education Forum/Global Engineering Deans Council (WEEF/GEDC)*, pp. 545–549, Nov. 2021, doi: <https://doi.org/10.1109/weef/gedc53299.2021.9657313>.
- [12] N. Kaur, M. Gupta, S. Saha, S. Manallack, and P. Gupta, "Financial Literacy through Gamification," Feb. 2024, doi: <https://doi.org/10.1109/iciptm59628.2024.10563365>.
- [13] R. S. Gautam, A. Rawal, and S. Rastogi, "Financial Literacy and Its Impact on Sustainable Development of India Using Internet Subscribers as Moderator," Oct. 2022, doi: <https://doi.org/10.1109/sibf56821.2022.9939760>.
- [14] S. Rani, "The Impact of Financial Literacy on Financial Well-Being: The Meditational Role of Personal Finance Management," Sep. 2023, doi: <https://doi.org/10.1109/ic3i59117.2023.10397970>.
- [15] R. Islam and I. Ahmed, "Gemini-the most powerful LLM: Myth or Truth," May 2024, doi: <https://doi.org/10.1109/ictc61510.2024.10602253>